

Aditya Mittha

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| Solapur, India

SUMMARY

Final-year Electronics and Telecommunication engineering student with hands-on experience building embedded systems around ESP32 and ARM-based microcontrollers. Proficient in Embedded C and Python with practical knowledge of communication protocols (UART, I2C, CAN, MQTT), interrupt-driven programming, and FreeRTOS task scheduling. Experienced in IoT product development — from sensor firmware to AWS IoT Core cloud integration. Comfortable debugging with logic analyzers, working on Linux environments, and collaborating with cross-functional teams.

SKILLS

Programming Languages: Embedded C, Python, Verilog, Bash/Shell Scripting
Microcontrollers & Hardware: ESP32, Raspberry Pi, Arduino, Logic Analyzer
Protocols & Interfaces: UART, I2C, CAN, MQTT (QoS), GPIO, ADC, Interrupts (ISR)
Cloud & IoT Platforms: AWS IoT, Firebase, MQTT Broker (Mosquitto), Git, GitHub
Soft Skills: Problem Solving, Team Collaboration, Self-Learning

EDUCATION

Bachelor's Degree in Electronics and Telecommunication	2027
Walchand Institute Of Technology	CGPA: 9.27
Higher Schooling	2023
D.H.B Soni College, Solapur	Percentage: 83%

WORK EXPERIENCE

Embedded System Intern	May 2025 – June 2025
Codec Technologies India	Remote
- Configured and tested UART and I²C communication interfaces on microcontrollers, writing Embedded C code to establish reliable data transmission between modules. - Identified and resolved hardware interfacing defects across embedded modules before production testing. - Managed version control with Git/GitHub across 2 active branches, maintaining project documentation and development workflows.	
Technologies: Embedded C, UART, I ² C, GPIO, Git, GitHub	

PROJECTS

AQUANOVA – Smart Water Pressure Management for Equitable Water Supply	2026
- Wrote ESP32 firmware to poll pressure and flow sensors over ADC/GPIO, publishing readings via MQTT (QoS) to Firebase cloud storage for real-time pipeline monitoring. - Built a Python analytics pipeline with pandas, numpy, and scikit-learn's Isolation Forest — detected leaks and pressure anomalies, reducing estimated maintenance response time.	
Role: IoT Firmware & Cloud Integration (3-person team)	
Tech Stack: ESP32, MQTT (QoS), Firebase, ADC, Pressure/Flow Sensors, Python, Telegram/SMS alerts	
SORTIFY – Waste Segregation System	2025
- Designed an AI-driven waste sorting system combining computer vision, weight sensing, and vibration-based sorting to classify household waste into 4 recyclable categories. - Integrated Raspberry Pi with camera module, weight sensor, and servo motor; automated identification, measurement, and bin control using Python and C. - Built a IoT dashboard tracking real-time segregation statistics, giving municipal teams actionable waste management data.	

Role: Embedded Hardware & Raspberry Pi Integration (Hackathon, 4-person team)
Tech Stack: Raspberry Pi, Embedded C, Python, Camera Module, HX711, Servo/PWM, Firebase, IoT Cloud

REWARDS & ACHIEVEMENTS

- 2x Winner, National Electronic Quiz Competition (Quizotronics)
- Runner-up, State-level chess championship
- Participated in 4+ Hackathons

LANGUAGES

English | Hindi | Marathi | Kannada